

Plug-and-Play Whole-Body Contact Estimation: A Controller-Agnostic Proprioceptive Force Estimator for Humanoids

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Abstract—A humanoid can estimate external contact forces over its whole body from proprioception alone, without force/torque sensors. But such estimators are trained end-to-end under *one* controller, whereas a deployed humanoid runs *many* — walking, several motion-tracking policies, compliant-interaction policies. A contact estimator would be far more useful as a *plug-and-play* module that drops onto any of them unchanged. We show the standard recipe does not transfer: trained on one controller and evaluated across seven others (kp/kd-scaled variants under an identical ground-truth force protocol), worst-case region localization falls to 0.42 and detection precision collapses to 0.53 — a different torque baseline is read as contact. We trace this to the estimator leaning on the raw actuator-torque channel, whose distribution the controller sets, and introduce a *controller-invariant residual channel*: a per-joint external-torque estimate from the equation of motion, depending only on the measured motion and the known model. It nearly eliminates the gap (worst-case region accuracy $0.42 \rightarrow 0.91$, precision $0.53 \rightarrow 0.97$ across all seven controllers). The same channel unexpectedly solves the hardest localization case: because it subtracts the ground reaction it removes double-support *stance masking*, raising leg-region recall from 0.56 to 0.89. Finally we make the channel hardware-realizable: the only simulation-privileged term is the ground reaction; dropping it leaves the upper body fully controller-invariant from proprioception alone (0.94), and supplying the ground reaction from a foot force estimate recovers the legs, degrading gracefully with estimation error (0.79 at 25% error). All experiments are in simulation on a Unitree G1.

I. INTRODUCTION

Physical interaction is central to what we want humanoids to do: leaning on a wall, being pushed, carrying a load, working in clutter. Detecting and localizing external contact is a prerequisite, and while dedicated skin or force/torque (F/T) sensing remains rare on full humanoids, recent results show that a surprising amount of contact information is recoverable from *proprioception alone* — joint positions, velocities, actuator torques, and IMU. SixthSense [1], for example, regresses a per-link wrench field over the whole body of a Unitree G1 from a short proprioceptive history.

These estimators are learned end-to-end, and crucially they are trained on data collected under *one* controller — the policy driving the robot during collection. In practice a humanoid is not one controller: it runs a walking policy, one or more

motion-tracking policies, a compliant-interaction policy [2], and so on. A contact estimator would be most valuable as an *infrastructure* component — a plug-and-play module that reads only proprioception and drops onto *any* of these controllers without re-collecting data and re-training (Fig. 1). Whether current estimators have this property has, to our knowledge, never been tested: the estimator and its single controller are always evaluated together.

This paper isolates and answers that question, and finds that the answer changes what such an estimator should take as input. Our contributions are:

- 1) We define and measure the *cross-controller generalization gap*. Using seven PD-gain-scaled controller variants under an identical ground-truth contact protocol (the controller is the only variable), a leave-one-out study shows the gap is large: worst-case region localization 0.42 and detection precision 0.53 on unseen controllers (Sec. III).
- 2) We identify the cause — the estimator leans on the raw actuator-torque channel, whose distribution the controller sets — and introduce a *controller-invariant residual channel* from the equation of motion. It nearly closes the gap (worst-case region accuracy 0.91, precision 0.97), robustly across all seven controllers including extreme extrapolation (Sec. IV, V-A).
- 3) We show the same channel *solves the legs*: subtracting the ground reaction removes double-support stance masking, raising leg-region recall $0.56 \rightarrow 0.89$, near arm level (Sec. V-B).
- 4) We make the channel *hardware-realizable*: its only simulation-privileged term is the ground reaction. Without it the upper body is still fully controller-invariant from proprioception alone (0.94); with a foot force estimate the legs are recovered and degrade gracefully with estimation error (Sec. V-C).

We are explicit about what a simulation study establishes (Sec. VI): “different controller” is emulated by gain scaling, not by genuinely different networks, so our gap is a *lower bound*.

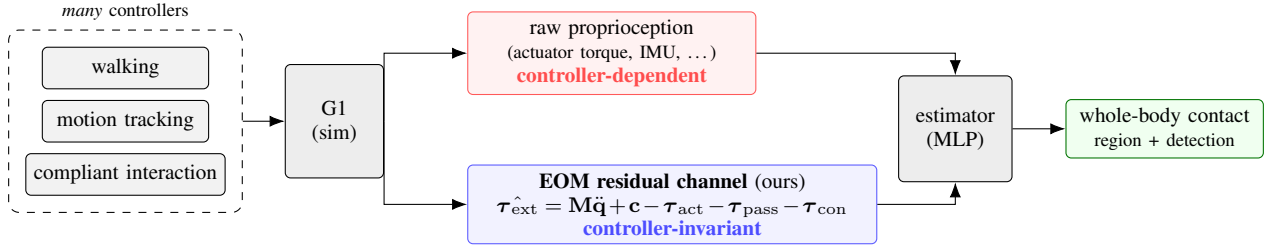


Fig. 1. **Plug-and-play contact estimation.** A humanoid runs many controllers, but a proprioceptive contact estimator is trained under one. The raw proprioception an end-to-end estimator relies on is **controller-dependent** (its actuator-torque distribution shifts with the control law), so the estimator does not transfer. We add a **controller-invariant** residual channel recovered from the equation of motion — it depends only on measured motion and the model, not on how the torque was produced — which lets one estimator plug into unseen controllers and, by subtracting the ground reaction, also un-masks the legs.

II. RELATED WORK

Proprioceptive contact/force estimation. SixthSense [1] estimates a whole-body per-link wrench field on a real G1 from a ~ 1 s proprioceptive window, and is the closest system to ours; it targets transient pushes and is trained and evaluated within a single control pipeline. Classical *momentum observers* [3] estimate external joint torques from generalized momentum without joint accelerations, independent of the control law by construction; MOB-Net [4] learns a floating-base momentum observer for whole-body wrench estimation, and FACTR 2 [5] self-supervises a free-space external-torque model for label-free, sensorless force estimation. Our residual channel is a learning *input* in the same spirit — a physics-derived, controller-invariant signal — rather than a standalone estimator, and we study specifically its effect on *cross-controller* transfer, and its side effect on leg observability.

Compliant humanoid interaction. GentleHumanoid [2] learns upper-body compliance for contact-rich interaction using an impedance formulation [6] and proprioception-only deployment; it solves how to *respond* softly to contact but does not localize it. A plug-and-play estimator is a natural complement: it would let such a controller know *where* it is being touched.

Randomization for transfer. Domain/dynamics randomization [7] trains across a distribution of environment parameters to obtain robustness to the sim-to-real gap. We use the analogous idea along a new axis — the *downstream controller* — and compare it against, and combine it with, the physics-based residual channel.

III. THE CROSS-CONTROLLER GENERALIZATION GAP

A. Setup

We use a Unitree G1 in MuJoCo [8], driven by a fixed, exported standing/motion-tracking policy through a position-PD actuator, in a sim-to-sim pipeline that reproduces the training-time observation and action semantics. External forces follow a scripted protocol (rest / ramp / hold / release, $|\mathbf{F}_{\text{ext}}| \in [10, 40]$ N, isotropic direction, one of 24 candidate links per cycle) giving exact ground-truth contact location and force; a frame is labeled *contact* when $|\mathbf{F}_{\text{ext}}| > 5$ N. The estimator reads a 6-step proprioceptive window (applied torque, joint-position history, angular velocity, projected gravity, past

TABLE I
THE SEVEN CONTROLLER VARIANTS (PD-GAIN SCALE VS. NOMINAL).

	base	soft	stiff	soft-ud	v-stiff	under-d	over-d
k_p	1.0	0.6	1.5	0.4	2.0	1.0	0.7
k_d	1.0	0.8	1.2	0.7	1.5	0.5	1.5

actions; 320 dims/frame) and has two heads: a detection logit and a 5-region classifier (left/right arm, left/right leg, trunk). We report detection precision/recall/F1 and region-localization accuracy on contact frames (regAcc).

Emulating different controllers. To vary the controller while holding everything else fixed, we scale the deploy PD gains, which enter the actuator torque $\tau = k_p(\mathbf{q}_{\text{des}} - \mathbf{q}) - k_d\dot{\mathbf{q}}$ directly — exactly the channel a different downstream policy would shift. We use **seven** controllers spanning $k_p \in [0.4, 2.0] \times$ and $k_d \in [0.5, 1.5] \times$ nominal, independently varied (Table I), and collect 360k frames each.

B. Result: the gap is large

Table II (“proprioception” rows) and Fig. 2 report a leave-one-out study: train on six controllers, evaluate on the held-out unseen one. Even with this randomization, raw proprioception is fragile — worst-case region accuracy 0.42 and detection precision 0.53 (the extreme-soft controller, an extrapolation below the training range), and only 0.65 mean accuracy. Under a different torque baseline the estimator reads contact-free stance as contact. A naive end-to-end estimator therefore does *not* plug and play, and because gain scaling is a comparatively small perturbation of the torque distribution, the gap for a genuinely different policy can only be larger.

IV. A CONTROLLER-INVARIANT RESIDUAL CHANNEL

The failure above is shortcut learning: the raw actuator torque is the most informative single channel for contact, but its distribution is set by the controller, so a model that leans on it overfits the controller. We supply instead a channel that carries the same contact information but is invariant to the control law.

The rigid-body equation of motion is

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) = \tau_{\text{act}} + \tau_{\text{passive}} + \tau_{\text{con}} + \tau_{\text{ext}}, \quad (1)$$

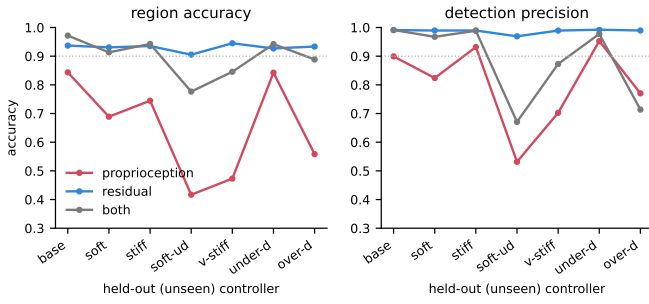


Fig. 2. Cross-controller transfer (leave-one-out over 7 controllers). Raw proprioception collapses on unseen/extreme controllers; the controller-invariant residual channel stays flat.

where $\mathbf{c}$ collects Coriolis, centrifugal and gravity terms, $\boldsymbol{\tau}_{\text{act}}$ is the actuator torque, $\boldsymbol{\tau}_{\text{con}} = \mathbf{J}_c^T \mathbf{f}_c$ is the constraint (ground-reaction) force, and $\boldsymbol{\tau}_{\text{ext}} = \mathbf{J}^T \mathbf{F}_{\text{ext}}$ is the external generalized force we care about. Solving for the external term,

$$\hat{\boldsymbol{\tau}}_{\text{ext}} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) - \boldsymbol{\tau}_{\text{act}} - \boldsymbol{\tau}_{\text{passive}} - \boldsymbol{\tau}_{\text{con}}, \quad (2)$$

gives, on the 29 actuated joints, an estimate of the external joint torque that depends only on measured motion ($\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}$), the measured actuator torque, and the known model — *not* on how $\boldsymbol{\tau}_{\text{act}}$ was chosen. We feed this 29-dim vector, windowed over 6 steps, as an additional input channel. Empirically $\|\hat{\boldsymbol{\tau}}_{\text{ext}}\| \approx 0$ in contact-free stance regardless of the gains and grows with the applied force, exactly the invariance we want. We evaluate three input configurations: *proprioception* (the 320-dim raw window; the baseline), *residual* (the 29-dim channel from Eq. 2 alone), and *both* (concatenated).

V. EXPERIMENTS

A. The residual channel closes the gap robustly

Table II gives the central result over the seven-controller leave-one-out study. The residual channel is the robust winner: worst-case region accuracy 0.91 and mean 0.93, with detection precision 0.97–0.99 on *every* held-out controller including the extreme-soft and extreme-stiff extrapolations, where raw proprioception falls to 0.42/0.47. The “both” configuration is best *in-domain* (e.g. 0.97 on central controllers, as it has the richest signal) but partially re-inherits the proprioception shortcut on extreme controllers (worst-case 0.78, precision 0.67), confirming that the raw torque channel is what breaks transfer. Controller randomization helps every input (it lifts proprioception from single-controller 0.40 to 0.70), but even with six training controllers raw proprioception trails the residual, which is essentially unchanged by randomization — the physics channel, not a bigger training set, is what buys plug-and-play.

B. The residual also solves the legs

Leg contacts are the classic hard case for proprioceptive estimation: in double support the feet are planted, and an external leg force is partly absorbed by the ground reaction, *masking* it in the raw joint signals. Breaking down per-region

TABLE II
CROSS-CONTROLLER ROBUSTNESS (LEAVE-ONE-OUT OVER 7 CONTROLLERS, TRAIN ON 6 / EVAL ON THE UNSEEN ONE). WORST-CASE AND MEAN OVER THE 7 HELD-OUT CONTROLLERS.

input	worst regAcc	mean regAcc	worst prec	mean prec
proprioception	0.417	0.653	0.53	0.80
residual (ours)	0.905	0.931	0.97	0.99
both	0.777	0.897	0.67	0.88

cross-controller robustness (7 held-out)

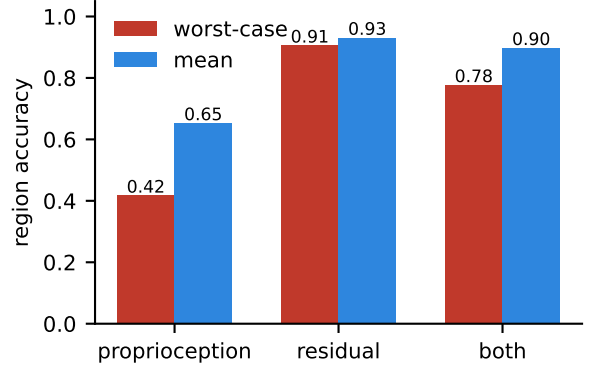


Fig. 3. Worst-case vs. mean cross-controller region accuracy over 7 held-out controllers. The residual channel raises the worst case from 0.42 to 0.91.

recall (Table III, Fig. 4) shows raw proprioception localizes arms well (0.80) but legs poorly (0.56) and trunk worse (0.43). The residual channel raises legs to 0.89 — near arm level — and trunk to 0.86. The reason is structural: Eq. 2 explicitly subtracts the ground reaction $\boldsymbol{\tau}_{\text{con}}$, removing exactly the stance-masking term that hides leg forces. A single physics channel thus addresses both plug-and-play transfer and leg observability.

C. Making the residual hardware-realizable

Of the terms in Eq. 2, the mass matrix, bias, actuator torque and passive forces are all computable on hardware from encoders, a model, and motor currents; the *only* simulation-privileged term is the ground reaction $\boldsymbol{\tau}_{\text{con}}$. We quantify the reliance on it in two steps.

Without any ground-reaction term (drop $\boldsymbol{\tau}_{\text{con}}$ entirely; fully proprioception-realizable), the residual on contact-free links is unchanged, so the *arms* remain fully controller-invariant at 0.94 (Table IV); the legs and trunk, now contaminated by the unremoved ground reaction, fall to 0.62 and 0.42. The upper-body plug-and-play claim is thus realizable with *no* extra sensing.

With a foot ground-reaction estimate — what a foot F/T sensor or a floating-base momentum observer [3], [4] provides — the legs are recovered. We model an imperfect estimate as $\hat{\boldsymbol{\tau}}_{\text{con}} = \boldsymbol{\tau}_{\text{con}} + \alpha \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_{\text{GRF}})$, and sweep the relative error α (Fig. 5). At $\alpha = 0$ (perfect estimate) legs return to 0.92; degradation is graceful — 0.84 at 10% error, 0.79 at 25%, 0.73 at 50% — with the arms unaffected throughout

TABLE III
PER-REGION RECALL (CROSS-CONTROLLER, 7-FOLD LEAVE-ONE-OUT AVERAGE).

input	arms	legs	trunk
proprioception	0.80	0.56	0.43
residual (ours)	0.98	0.89	0.86

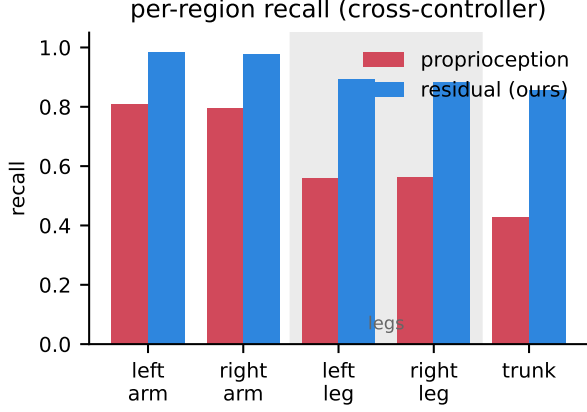


Fig. 4. Per-region recall. The residual channel rescues the legs (and trunk) by removing stance masking, closing the gap to the arms.

(0.97). Since a foot F/T sensor typically has well under 10% error, the leg result is realizable on any humanoid with foot force sensing, and robust to its noise; the trunk is the most estimate-sensitive region.

VI. DISCUSSION AND LIMITATIONS

“Different controller” is emulated by gain scaling. We did not train several distinct networks; we scaled PD gains, which shifts the actuator-torque distribution — the channel that matters — but is a smaller perturbation than a genuinely different policy. Our measured gap is therefore a *lower bound*, and confirming that the residual transfers across truly different trained controllers is the necessary next step.

Ground reaction on hardware. The full residual reads τ_{con} from the simulator; on hardware it comes from a foot F/T sensor or a floating-base momentum observer [3], [4]. Sec. V-C bounds the impact: the upper body needs nothing extra, and the legs tolerate substantial ground-reaction error, but a hardware demonstration remains future work.

Scope. All results are in simulation, at 5-region granularity, with sustained forces; we have not evaluated per-link wrench regression (matching SixthSense’s output), force-magnitude accuracy, walking/dynamic tracking, or a real robot. These are the axes to widen before claims of practical deployment.

VII. CONCLUSION

We asked whether a proprioceptive whole-body contact estimator can serve as a plug-and-play module across the many controllers a humanoid runs. The standard end-to-end recipe does not transfer — a different controller shifts the

TABLE IV
REALIZABILITY OF THE RESIDUAL. “FULL” SUBTRACTS THE (PRIVILEGED) GROUND REACTION; “REALIZABLE” DROPS IT (PROPRIOCEPTION-ONLY).

residual	arms	legs	trunk	worst-case regAcc
full (privileged GRF)	0.98	0.89	0.86	0.905
realizable (no GRF)	0.94	0.62	0.42	0.586

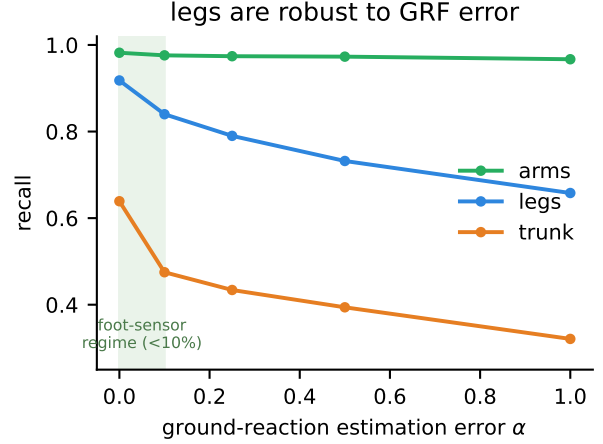


Fig. 5. Legs are robust to ground-reaction estimation error: recall degrades gracefully with α while the arms are unaffected. A realistic foot-sensor error ($< 10\%$) keeps legs near 0.86–0.89.

actuator-torque distribution the estimator depends on, collapsing accuracy and precision. A controller-invariant residual channel, recovered from the equation of motion, restores transfer robustly across a wide range of controllers, and as a by-product removes the double-support stance masking that makes legs hard, raising leg recall to arm level. The channel is hardware-realizable: the upper body needs only proprioception, and the legs need only a foot ground-reaction estimate, to which they are robust. A genuine plug-and-play contact-sensing module for humanoids therefore runs not through a bigger passive estimator, but through the right, controller-agnostic representation of the external force.

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